Heavy-Copper & Power PCBs

4 oz / 6 oz / 10+ oz copper for EV inverters, motor drives, solar

When 1-oz isn't enough — into the world of power PCB

Audience: Power electronics engineers

Heavy Cu • Bus bars • Thermal • Etching limits • Specialty fabs

What is Heavy Copper

- Heavy copper PCB: copper weight ≥ 4 oz/ft² (140 µm thick)
- Standard PCB: 1 oz typical (35 µm)
- Carries 10-200 A per trace depending on width + temp rise
- Used in: EV inverters, solar combiners, motor drives, welding gear, train traction
- Cost: 2-5× standard PCB
- Available copper weights: 4, 6, 8, 10, 15, 20, 30 oz
- At extreme: bus bar PCBs combine PCB + solid Cu bus bars

Heavy Copper Specs

Cu weight	Thickness	Min trace width	Min spacing	Annular ring
4 oz	140 µm	0.38 mm (15 mil)	0.38 mm	0.38 mm
6 oz	210 µm	0.50 mm (20 mil)	0.50 mm	0.50 mm
10 oz	350 µm	0.76 mm (30 mil)	0.76 mm	0.76 mm
15 oz	525 µm	1.0 mm (40 mil)	1.0 mm	1.0 mm
20 oz	700 µm	1.5 mm (60 mil)	1.5 mm	1.5 mm
30 oz	1050 µm	2.5 mm (100 mil)	2.5 mm	2.5 mm

Why coarser at heavier weight

Etching thicker copper takes longer → wider undercut → minimum trace + spacing grow.

Heavy Copper Applications

EV inverters

200-400 A bus connections
6-10 oz typical

Solar combiners

30-100 A DC strings
3-4 oz typical

Motor drives

10-50 A BLDC drive
2-4 oz

Welding power

100-200 A DC
6+ oz

Battery packs

Cell interconnects, BMS
4-10 oz

UPS / Telecom

Battery + rectifier paths
3-4 oz

Manufacturing Challenges

- Etching: thicker Cu = longer etch time → wider undercut → coarser features
- Drilling: harder to drill clean through 10+ oz Cu
- Plating: thicker Cu in PTH walls needs longer plating cycle
- Lamination: thicker stacks need higher pressure + more vacuum
- Profile: heavy Cu PCBs are HEAVY — handling considerations
- Solder: regular reflow profile may not work — thicker Cu needs higher peak temp
- Soldermask: thicker Cu means thicker mask needed → adjustments to standard process
- Most fabs cap at 2-3 oz outer; specialty fabs handle 4-30 oz

Mixed Cu Weight Stackups

- Common: 2-4 oz on outer signal/power layers; 1-2 oz on inner signal layers
- Hybrid stackup: 4 oz power outer + 0.5 oz signal inner
- Allows compact, high-current design without making whole stackup heavy
- Specify per layer: 'L1: 2 oz, L2: 0.5 oz, L3: 0.5 oz, L4: 2 oz'
- Lead time + cost premium ~1.3-1.8× standard
- Most fabs (JLCPCB included) support mixed weights up to 2 oz; for 4 oz+ need specialty fab
- Impedance-controlled traces: width recalculated for mixed weights

Bus Bar PCBs

- Bus bar = solid copper conductor mounted to PCB for ultra-high current
- Used for: 200+ A continuous, EV motor drives, large welding/inverter systems
- Cu cross-section: 5×10 mm to 20×50 mm typical
- Mounted via: threaded holes, screwed-on lugs, embedded into PCB
- Embedded bus bar: solid Cu laminated INSIDE the PCB stackup
- Cost: significantly more than heavy-Cu PCB
- Used by: high-end EV inverter manufacturers (Bosch, Vitesco)

Thermal Considerations

- Heavy Cu = excellent thermal conductor (better heat-spreader than 1 oz)
- Combine with thermal vias for IC heat sinking
- Heat-sink bonded directly to heavy-Cu PCB (mech screws + thermal compound)
- For very-high-current paths: thermal pad of IC must be sized to match Cu pour
- Consider Cu thickness in IR drop calculations (heavy Cu = lower drop = less heat)
- Use thermal camera at full load — find hot spots
- Heavy Cu doesn't help if THERMAL PATH is blocked (e.g., by enclosure)

Design Rules for Heavy Cu

- Trace width: per IPC-2152 (or fab spec) — generous for thermal margin
- Trace spacing: same as min trace width minimum
- Annular ring: same as min spacing
- Via diameter: increase 1.5× over standard for current-handling
- Plane stitching: extra-dense for current return + thermal
- Avoid sharp corners on power traces (use curves or 45°)
- Component pad sizes: larger than standard for thermal mass
- Soldermask opening: per fab spec for heavy Cu (slightly larger)

Specialty Fab Sources

- International: PCB Universe, Multech, Sierra Circuits
- Asia: Wurth Elektronik, AT&S, Ibiden
- India: AT&S India, Centum, specialty PCB fabs in Pune/Bengaluru
- Talk to your CM about heavy-Cu capability — many won't accept > 2 oz
- Lead time: 2-4 weeks typical for heavy Cu (vs 5-10 days standard)
- Cost: 3-5× standard, depending on Cu weight
- Get multiple quotes — pricing varies significantly

Common Heavy Cu Mistakes

- Specifying heavy Cu without confirming fab capability
- Using standard 6-mil trace rules on 4-oz Cu → fab can't etch reliably
- Underestimating thermal — heavy Cu sinks heat → soldering profile may need adjustment
- Not accounting for IR drop on long traces — thicker but not infinite Cu
- Mixing heavy + thin Cu without confirming fab supports it
- Skipping bus bar consideration → designing impossible-density board
- Reflow profile too cool for thick Cu solder pads → cold joints
- Forgetting heavy Cu PCBs are HEAVY — handling stress in test fixtures

Key Takeaways

Key takeaways from this lecture

4-30 oz spec

Pick by current demand.

Coarser features

Etching wider than thin Cu.

Mixed weight common

Heavy outer, thin inner.

Specialty fab needed

Most CMs cap at 2 oz outer.

Bus bars for 200 A+

PCB + solid Cu.

Thermal benefit

Heavy Cu spreads heat.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- IPC-2152 — current carrying capacity
- Heavy copper PCB vendor white papers
- AT&S / Wurth heavy Cu spec sheets
- Phil's Lab — heavy Cu walkthrough
- Bosch / Vitesco EV inverter publications
- PCB Universe + Sierra Circuits design e-books

Thank you • Build something this week.